INTERNAL TECHNICAL & SYSTEM ANALYSIS DISCUSSION PAPER

10 MW / 40 MWh (0.25C, 4-hour) Battery Energy Storage System (BESS)

Audience: System Planning, T&D;, Operations, Protection
Status: Internal engineering paper for decision support (source-checked)

1. Executive Summary

This paper consolidates engineering positions for a 10 MW / 40 MWh (0.25C, 4-hour) grid-connected BESS. It formalizes control modes, quantitative performance targets, model requirements, and verification steps. PQ and droop modes are mandatory for grid-connected operation; VF (grid-forming) is optional/beneficial and deployed selectively based on system studies. Targets reflect international practice and proven deployments of grid-forming BESS (Hornsdale VMM; ESCRI-SA) and microgrids (Snohomish Arlington).

2. System Context & Objectives

The grid is integrating higher shares of inverter-based resources (IBR). The 10 MW BESS is expected to: (i) provide primary frequency response; (ii) supply reactive support and voltage regulation at the point of connection (POC); (iii) ride through credible faults and disturbances; and (iv) avoid adverse control interactions with existing IBR and synchronous plant. Day-to-day use emphasises grid-following operation with droop-based support. Grid-forming operation (VF) is considered an option for weak/critical nodes or where resilience objectives justify added complexity.

3. Control Architecture & Operating Modes

Mode	Control Objective	Status / Deployment Guidance
VF (grid-forming)	Regulate bus V and f; controlled short-circuit contribution; fast AVR as offered	Optional / Beneficial — apply at weak/critical nodes o
PQ (grid-following)	Track P*, Q* at POC under PLL synchronization	Mandatory — default day-to-day operating mode
Droop (P–f & Q–V)	Provide primary frequency and voltage/reactive support; enable load sharing	Mandatory — required for stability and local support

4. Quantitative Performance Targets (POC/SCR-tuned)

Function	Requirement / Target (internal; aligned to Appendices numerical standards)
P–f droop (primary frequency)	Droop 1–9% (4% default); configurable deadband; response ≤ 0.2 s
Q–V droop / VAR control	Stable Q–V droop; V, Q, or PF modes at POC; tune with documented stability mar
Frequency withstand	Continuous operation across utility 50■Hz bands used internally; avoid nuisance tr
RoCoF tolerance	Retain synchronism for ramps up to 4 Hz/s (250 ms avg) and 2.5 Hz/s (500 ms) in
LVRT/HVRT ride■through	Ride■through per internal voltage bands; reactive current injection during dips; act
Reactive current injection (faults)	Provide ~2% reactive current for each 1% drop in retained voltage during faults (≈2
Set■point timing	Start ≤ 50 ms; achieve ≤ 300 ms; settle ≤ 400 ms (non■oscillatory)

Function	Requirement / Target (internal; aligned to Appendices numerical standards)
Power quality	Harmonics, flicker, unbalance within specified limits at POC
Reactive capability	≥ ±0.3 · Pn at POC (four■quadrant)
Over■current envelope	Declare OEM■validated continuous and short■time current limits; claims above ~

5. POC Grid Strength & SCR Criteria

Operating Mode / Study Case	Minimum Grid Strength Target & Notes
Grid■following (PQ + droop, default)	Design/tune for robust operation at SCR ≥ ~3; validate small■signal margins and
Grid■forming (VF, optional)	Demonstrate stable, controllable operation at minimum SCR ≈ 1.0 with current limi
Mixed IBR environments	Verify no adverse interactions; if POD is offered, demonstrate damping of 0.2–2.5

6. Formal Control Definitions

- P-f droop: ΔP = –Kpf · Δf; Kpf from droop (%) and Pn; include rate limiting & anti-windup.
- Q-V droop: ΔQ = +Kqv · ΔV; define reactive priority; demonstrate small- and large-signal stability margins.
- VF option: Outer V,f regulators with inner current limiting; controlled short-circuit contribution; fast AVR where implemented.
- PLL & synchronization (GFL): robust under low inertia; declare lock range, RoCoF tracking, and recovery dynamics.
- Limiter coordination: coordinate P/Q limiters with priority to avoid wind-up and mode chattering.

7. Protection & Fault Behaviour (POC Coordination)

- Fault current: declare peak/AC components and duration with limiting strategy; coordinate with breaker ratings.
- Trip logic: trip only to protect equipment integrity; otherwise ride-through; document thresholds.
- Communications loss: define fail-safe (hold last / set-point ramp-down) and recovery logic.

8. RMS & EMT Models - Requirements

- Provide RMS (phasor) and EMT models with functionally equivalent behaviour; declare tool versions and solver/time-steps (typical: RMS 5-10 ms; EMT 25-50 µs).
- Include control blocks: P-f droop; Q-V droop/AVR; LVRT/HVRT; reactive current injection; PLL. Optional: virtual inertia, POD, VF↔GFL seamless transition.
- Parameter spreadsheets (units, min/max, tuned values); include current limiters and thermal protection tied to the over-current envelope.
- Mandatory benches: frequency step (±0.2 Hz); voltage step (±5%); LVRT/HVRT sequences; RoCoF ramps; near/far faults; optional POD injection.
- Equivalence evidence: overlay RMS vs EMT for identical cases; provide run scripts/projects and plots.

9. Verification & Acceptance

- Type/FAT: certified reports for droop timing, voltage/reactive support, ride-through and recovery timings, and power quality spectra.
- SAT: witnessed tests for set-point timing, LVRT/HVRT sequences, reactive injection and recovery, RoCoF tolerance; verify SCADA/AGC/AVC interoperability.
- Non-conformance: vendor to correct and resubmit models/firmware with change logs and comparative plots.

Integrated Requirements (from uploaded specification)

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1. Executive Summary

This paper consolidates engineering positions for a 10 MW / 40 MWh (0.25C, 4-hour) grid-connected BESS. It establishes control modes, quantitative performance targets, model requirements, and verification steps. PQ and droop modes are mandatory for grid-connected operation; VF (grid-forming) is optional/beneficial and deployed selectively based on system studies. Targets reflect international practice (IEEE 1547 family) and proven deployments of grid-forming BESS and microgrids.

2. System Context & Objectives

The grid is integrating higher shares of inverter-based resources (IBR). The 10 MW BESS is expected to:

- Provide primary frequency response
- Supply reactive support and voltage regulation at the point of connection (POC)
- Ride through credible faults and disturbances
- Avoid adverse control interactions with existing IBR and synchronous plant

Day-to-day use emphasizes grid-following operation with droop-based support. Grid-forming operation (VF) is considered an option for weak/critical nodes or where resilience objectives justify added complexity.

3. Control Architecture & Operating Modes

The Power Conversion System (PCS) implements inner current control with outer power/frequency and voltage/reactive loops, supervised by coordinated limiters and protection. Three canonical operating modalities are defined:

Mode	Control Objective	Status / Deployment Guidance
VF (grid-forming)	Regulate bus V and f; controlled short-circuit contribution; fast AVR as offered	Optional/Beneficial - apply at weak/critical nodes or when studies show net stability/resilience benefit
PQ (grid-following)	Track P*, Q* at POC under PLL synchronization	Mandatory - default day-to-day operating mode
Droop (P-f & Q-V)	Provide primary frequency and voltage/reactive support; enable load-sharing	Mandatory - required for stability and local support

4. Quantitative Performance Targets

4.1 Frequency Response Requirements

Function	Requirement / Target
P-f droop (primary frequency)	Droop 1-9% (4% default); configurable deadband; response ≤ 0.2 s
Frequency withstand	Continuous operation across utility 50-Hz bands; avoid nuisance tripping
RoCoF tolerance	Retain synchronism for ramps up to 4 Hz/s (250 ms avg) and 2.5 Hz/s (500 ms avg)

Based on Sri Lankan Grid Code requirements, the BESS shall operate within frequency bands:

- **Normal Operation:** 49.50 Hz to 50.50 Hz (±1% of nominal 50 Hz)
- **Emergency Conditions:** 47.0 Hz to 52.0 Hz
- **Critical Protection Thresholds:** Above 51.20 Hz and below 48.75 Hz (ALFC suspension points)

4.2 Voltage and Reactive Power Control

Function	Requirement / Target
Q-V droop / VAR control	Stable Q-V droop; V, Q, or PF modes at POC; documented stability margins
LVRT/HVRT ride-through	Ride-through per voltage bands; reactive current injection during dips; active P recovery target ≥95% in ~100 ms post-clear
Reactive current injection (faults)	Provide ~2% reactive current for each 1% drop in retained voltage during faults (≈2 pu/pu k-factor)
Reactive capability	≥ ±0.3 · Pn at POC (four-quadrant)

4.3 Dynamic Performance

Function	Requirement / Target
Set-point timing	Start ≤ 50 ms; achieve ≤ 300 ms; settle ≤ 400 ms (non-oscillatory)
Power quality	Harmonics, flicker, unbalance within specified limits at POC
Over-current envelope	Declare validated continuous and short-time current limits; claims above ~120% for ≥60-120 s require thermal/oversizing justification

5. POC Grid Strength & SCR Criteria

Screen the point of connection (POC) short-circuit ratio (SCR):

Operating Mode / Study Case	Minimum Grid Strength Target & Notes
Grid-following (PQ + droop, default)	Design/tune for robust operation at SCR ≥ ~3; validate small-signal margins and PLL bandwidth limits
Grid-forming (VF, optional)	Demonstrate stable, controllable operation at minimum SCR ≈ 1.0 with current limiting; show seamless VF↔GFL transition (if offered)
Mixed IBR environments	Verify no adverse interactions; if Power Oscillation Damping (POD) is offered, demonstrate damping of 0.2-2.5 Hz modes

6. Formal Control Definitions

6.1 Primary Control Functions

- **P-f droop:** ΔP = –Kpf · Δf; Kpf from droop (%) and Pn; include rate limiting & anti-windup

- **Q-V droop:** $\Delta Q = +Kqv \cdot \Delta V$; define reactive priority; demonstrate small- and large-signal stability margins

6.2 Grid-Forming Option (VF)

- Outer V,f regulators with inner current limiting
- Controlled short-circuit contribution
- Fast Automatic Voltage Regulation (AVR) where implemented

6.3 Synchronization (Grid-Following)

- **PLL & synchronization (GFL):** Robust under low inertia; declare lock range, RoCoF tracking, and recovery dynamics
- **Limiter coordination:** Coordinate P/Q limiters with priority to avoid wind-up and mode chattering

7. Protection & Fault Behaviour

7.1 Fault Current Management

- **Fault current:** Declare peak/AC components and duration with limiting strategy; coordinate with breaker ratings
- **Trip logic:** Trip only to protect equipment integrity; otherwise ride-through; document thresholds
- **Communications loss:** Define fail-safe (hold last / set-point ramp-down) and recovery logic

7.2 Compliance with Sri Lankan Standards

Based on CEB Grid Code requirements:

- Comply with Limited Frequency Sensitive Mode (LFSM) requirements
- LFSM-O (over-frequency): Active power reduction for frequencies above threshold
- LFSM-U (under-frequency): Active power increase for frequencies below threshold
- Primary Frequency Regulation Mode with droop settings 2% to 9% (default 4%)

8. RMS & EMT Models - Requirements

8.1 Model Specifications

- Provide RMS (phasor) and EMT models with functionally equivalent behaviour
- Declare tool versions and solver/time-steps (typical: RMS 5-10 ms; EMT 25-50 µs)

8.2 Required Control Blocks

- P-f droop; Q-V droop/AVR; LVRT/HVRT; reactive current injection; PLL
- Optional: virtual inertia, POD, VF↔GFL seamless transition

8.3 Documentation Requirements

- Parameter spreadsheets (units, min/max, tuned values)
- Include current limiters and thermal protection tied to the over-current envelope

8.4 Mandatory Test Benches

- Frequency step (±0.2 Hz); voltage step (±5%)
- LVRT/HVRT sequences; RoCoF ramps; near/far faults
- Optional POD injection

8.5 Model Validation

- **Equivalence evidence:** Overlay RMS vs EMT for identical cases
- Provide run scripts/projects and plots

9. Verification & Acceptance

9.1 Type Testing and Factory Acceptance Testing (FAT)

- Certified reports for droop timing, voltage/reactive support, ride-through and recovery timings, and power quality spectra

- Compliance with relevant international standards (IEC 62619, IEC 62933, IEEE 1547)

9.2 Site Acceptance Testing (SAT)

- Witnessed tests for set-point timing, LVRT/HVRT sequences, reactive injection and recovery, RoCoF tolerance
- Verify SCADA/Automatic Generation Control (AGC)/Automatic Voltage Control (AVC) interoperability
- Demonstrate minimum system availability of 97% on monthly basis

9.3 Performance Monitoring

- 400 full equivalent cycles per year capability
- Minimum dispatchable capacity degradation curve:
- Year 1: 97.5% minimum
- Year 15: 62.5% minimum (based on CEB RFP requirements)

9.4 Non-Conformance Management

- Vendor to correct and resubmit models/firmware with change logs and comparative plots

10. Technical Standards Compliance

10.1 International Standards

- **IEEE 1547-2018:** Standard for Interconnection and Interoperability of Distributed Energy Resources
- **IEC 62933:** Electrical Energy Storage Systems
- **IEC 62619:** Secondary lithium batteries for the propulsion of electric road vehicles

10.2 Sri Lankan Grid Code Compliance

- CEB Grid Code (July 2024) requirements for Inverter Based Renewable Energy Technologies (IBRE)
- 33 kV connection level as per CEB RFP specifications
- Coordination with National System Control Centre (NSCC) dispatch protocols

10.3 Power Quality Standards

- Total Harmonic Distortion (THD) limits as per IEEE 519
- Voltage unbalance, flicker, and harmonic emission limits
- Electromagnetic compatibility requirements

11. Implementation Considerations

11.1 System Integration

- Coordination with existing Under Frequency Load Shedding (UFLS) schemes
- Integration with CEB's Automatic Load Frequency Control (ALFC) system
- SCADA connectivity to NSCC for real-time monitoring and control

11.2 Environmental and Safety

- NFPA 855 compliance for stationary energy storage systems
- UL 9540 and UL 9540A certification requirements
- Environmental impact assessment and mitigation measures

11.3 Operational Readiness

- Training programs for CEB operations personnel
- Maintenance protocols and spare parts management
- Emergency response procedures and safety protocols

Document Control:

- **Version:** 1.0
- **Date:** September 2025

- **Classification:** Internal Use
- **Review Cycle:** Annual or as required by system changes